

Sex Differentiation and the Social Organization of Science



In the last half of the 1960s, in response to interest in the position of women in American society, reports of the experience of female scientists began appearing in scientific journals (Rossi, 1965a, 1965b; Roe, 1966; Parish and Block, 1968; Graham, 1970; Maccoby, 1970; White, 1970). Drawing heavily on the authors' personal experience and observations, these papers introduced the status of women scientists as a subject for scientific investigation and proposed several hypotheses to account for their status. Empirical studies followed, and by 1975 researchers had accumulated considerable evidence for discrepancies in the positions of women and men in science (Simon, Clark, and Galway, 1967; Bayer and Astin, 1968; Fidell, 1970; Perrucci, 1970; Cole, 1971; Lewin and Duchan, 1971; Astin and Bayer, 1972; Reskin, 1973; Bayer and Astin, 1975) as well as in several individual disciplines (Morlock, 1973; Green,

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1976). The current state of the knowledge is summarized in two review essays (Zuckerman and Cole, 1975; Vetter, 1976) and in a review of the literature on sex differences among graduate students (Solmon, 1976).

During this period, several researchers considered whether the observed discrepancies resulted from sex discrimination, but the term *discrimination* rarely appeared in titles, and the debate was carried out circumspectly in the articles' concluding pages (see, for example, Cole and Cole, 1973; Zuckerman and Cole, 1975; Reskin, 1976). Recently some researchers have tried either to identify data sets (Cartter and Ruhter, 1975) or to develop statistical techniques (Reskin, 1977a) better suited to assessing sex discrimination, but much remains to be done on both tasks.

Thus, the study of sex differences in science has become a quasi-specialty, but work in this area has rarely been specific to science. Instead, it has depended primarily on more general theories developed to explain sex differences among academics or other professionals (Epstein, 1970a). A few studies, framed in terms of the normative structure of science, have explored the extent to which status attainment within science is universalistic (Cole and Cole, 1973; Reskin, 1976). In general, however, insights from the sociology of science have contributed little to this research area, and results have offered little in return to help explain the organization of the scientific community. This is not surprising. Political rather than theoretical considerations originally prompted these studies, and a certain amount of research was necessary just to determine the actual extent of sex differences.

Now we know more about the relative positions and performance of female and male scientists (Astin, 1977; Reskin, 1978), and it is time to consider how the organization of scientific work affects the presence, form, and extent of sex differentiation in science. For example, what aspects of the social organization of science foster sex differentiation? What aspects impede it? In what way are social patterns of sex-role differentiation reproduced in the scientific community? How does sex differentiation affect role performance?

In this chapter I consider some ways in which specific features of the scientific enterprise interact with social patterns of sex differentiation and stratification to introduce sex differentiation

into the scientific community. My discussion is necessarily speculative. We know too little about actual patterns of sex differentiation in collegial networks, patterns of collaboration, or the functional division of labor to make firm claims. At the very least, this chapter should illustrate the value of grounding future research on sex differences in our knowledge of the social organization of the scientific community.

Scientific Collegueship and Sex Differentiation in Science

Scientists advance knowledge, train future scientists, and administer organizations that fund or pursue research (Zuckerman and Merton, 1972, p. 314). Of these activities, teaching and administration are necessary for the practice of science, but without research science could not exist. To make research possible, the research role prescribes a set of technical behaviors and an orientation with which scientists approach their own work and that of others (Feigl, 1953, pp. 8–18; Merton, 1957; Storer, 1966, p. 81). Because research occurs in a social context, the role of researcher specifies certain reciprocal relations with other researchers. These may occur as public, formal exchanges, such as presentations at meetings or published reports and the responses they elicit (Hagstrom, 1965; Storer, 1966), or as the informal interaction that comprises most exchanges between scientific workers. These reciprocal relations include the collegial behaviors that are implicit in the research role, and they, too, are necessary for carrying out scientific research.¹

Some scientific roles—for instance, teacher and apprentice, project director and research associate—involve partners of unequal status. In contrast, the collegial role implies an egalitarian, symmetrical relationship between scientific researchers. The norms of science dictate some aspects of the collegial role. For example, communalism requires scientists to share their findings with co-workers, who in turn are obliged to evaluate them critically (Storer, 1966, p. 90). But scientists cannot follow even a fraction of the papers in their field (Price, 1963), much less respond competently to them all; so, realistically, collegial obligations are limited primarily to institutional and specialty colleagues, among whom personal contact is possible.

Other aspects of the collegial role have developed in response to the structure and needs of the scientific enterprise. The size and the complexity of modern science make it impossible for scientists to monitor all developments in their field, so they rely on colleagues to call to their attention references or results that they may have missed. Similarly, the high degree of specialization encourages scientists to draw upon colleagues for technical assistance, occasional instruction, and collaboration. The difficulty and uncertainty of research make scientists dependent on colleagues for advice and encouragement when they get hung up in their work; as potential competitors, colleagues provide the impetus for completing and publishing work. The reward system directs scientists to colleagues for appraisal, recognition, and criticism of their work. Because clearly defined standards of performance seldom exist, colleagues function as a reference group against whom scientists can evaluate their own performance. And, finally, the long hours associated with scientific work often limit scientists' opportunities to develop primary relationships away from their work, so they turn to colleagues for social companionship. In sum, the collegial role calls for scientists to teach, collaborate with, encourage, inform, evaluate and reward, compete with and befriend co-workers. Such reciprocal relations are essential for the pursuit of scientific inquiry.

Nothing in the collegial role inherently precludes collegiality between female and male co-workers. However, since the role applies to relationships between scientific researchers of approximately equal status, collegial responses usually do not evolve naturally between researchers of unequal status on either scientific or nonscientific dimensions. Instead, such co-workers usually relate to each other in terms of roles that do not assume equality. For example, few scientists treat their students and technicians as colleagues; their statuses do not elicit collegial responses, and the scientific roles of teacher-student (or administrator-employee) provide appropriate alternative models for interaction. Similarly, because women's lower gender status is inconsistent with the implicit status equivalence of colleagues, sex stratification itself can block normal collegiality between male and female researchers.

Extrascientific status distinctions also may hinder normal collegial relations in other ways. Certain "master status-

determining traits" (Hughes, 1945, p. 357; or simply "master statuses," in Kanter's, 1977, terms)—such as sex, race, and age—typically take precedence over other categories in dictating social responses. These statuses are usually elaborated in and supported by a variety of role relationships that specify modes for interaction between role partners who occupy different positions on the master status dimension. In the case of gender, such role relationships include wife-husband, mother-son, father-daughter, boyfriend-girlfriend, sister-brother, lovers, and so forth. Specific role requirements for each vary across social groups, and role partners have considerable latitude. But most Americans, including scientists (Mitroff, Jacob, and Moore, 1977, p. 323), endorse and conform to the sex roles appropriate to their social positions. Whereas collegial roles are symmetrical and egalitarian, sex roles are differentiated by gender and in most cases hierarchical, granting males greater power and privilege.

Thus, female and male co-workers are subject to conflicting, mutually exclusive role models for their relationships. At one extreme, they may ignore the roles dictated by their gender status and follow the traditional collegial model. At the other, they may relate to each other exclusively in terms of the nonscientific sex roles that most closely fit their scientific situation (for example, roles primarily dictated by gender etiquette that govern the relations between female and male *nonscientific* co-workers). A third alternative is minimizing their interaction. Several observers contend that the third alternative occurs frequently (Epstein, 1970a; White, 1970; Patterson, 1971, p. 233), and evidence that women are excluded from collegial networks (Bayer, 1970, p. 28; Kaufman, 1978) supports their claim. Perrucci (quoted in White, 1970, p. 414) notes that female scientists are more likely than males to stress the importance of working with colleagues and associating with scientists of recognized ability, probably because such opportunities are scarcer for them than for men. I argue, however, that male and female scientists often choose an intermediate option: they adapt conventional sex roles to the scientific setting to create a hybrid of gender and collegial roles that systematically introduces sex-role differentiation into the scientific community. Before elaborating on this argument, I should note the following qualifications.

First, the role of scientific colleague evolved when all scientists were men. The traditional masculinity of the collegial relationships probably is not sufficient to preclude cross-gender colleagueship, but it does reinforce the other barriers described above.²

Second, status inequality that originates outside the scientific community also can hinder the development of collegial relationships between other subgroup members. Any pronounced discrepancy in the status of co-workers could interfere with the development of a collegial relationship. If differentiated roles are associated with the status on which persons differ (as with gender status and sex roles), these may be substituted for collegial roles. For example, age, social class, and race are hierarchically differentiated, master status-determining characteristics, just as gender is; but all are not supported to the same degree by a system of status-specific roles. Age is most like gender in the existence of generally recognized roles that govern interaction between persons of different ages. Though few scientific researchers are extremely young, the youth of the rare prodigy, like the gender of a female scientist, is inconsistent with the collegial mode of interaction. Despite even prodigious accomplishments, the role responses elicited by youth are at odds with the collegial role: older scientists may behave paternalistically, while the young may respond with the deference traditionally due to older people. (See Zuckerman and Merton, 1972, for a discussion of age stratification in science.) Class and racial roles exist, but they are not universal in our society. It should not be surprising, then, that—at least with respect to class—the effects on scientific interaction are unclear (see Crane, 1969, 1976; Gaston, Wolinsky, and Bohleber, 1976).

Third, I do *not* claim that male scientists deliberately respond to female co-workers in terms of sex roles rather than collegial ones in order to hinder women's careers (although some may do so) or even that they unconsciously initiate such substitutions. Rather, the possibility of substitution inheres in the incompatibility of egalitarian collegial roles and stratified gender roles, and strict reliance on collegial roles can occur only when *both parties* treat gender as irrelevant. If scientists of either sex find a collegial relationship uncomfortable or inappropriate, they can abstain; without two willing role partners, colleagueship cannot occur. Status dis-

crepancies potentially complicate what otherwise would be routine interaction, and for this reason alone scientists of either sex might prefer sex roles to renegotiating the nature of their relationship for each encounter (Epstein, 1970b, p. 978). Moreover, scientists learn sex roles long before they learn how colleagues interact, and some scientists of both sexes might find sex roles easier to perform. Yet women stand to lose considerably more than men do from the incorporation of sex roles into scientific interaction. Because men constitute a substantial majority in most fields, male scientists have many potential colleagues. But women who choose to relate to male co-workers in terms of modified sex roles may forgo the possibility of true collegueship. In addition, because the content of sex roles reflects gender stratification, their use introduces both gender-based task differentiation and sex stratification into the scientific community.

Finally, some female scientists do become more integrated over time, as co-workers get used to their presence and recognize the value of their work (Pour-El, 1974, p. 37). Unfortunately, many women do not overcome their early collegial isolation and remain sufficiently productive to enjoy the ultimate rewards of collegial acceptance.

Models for Cross-Gender Relationships

If traditional sex roles provide models for relationships between female and male scientists, it follows that scientists must draw on a limited repertoire of models. Those models are based primarily on kinship or romantic relationships, although quasi-scientific superior-subordinate models (such as scientist and secretary or assistant) are also possible.

Models Based on Kinship Roles. Kinship roles provide scientists of both sexes with models for relating to co-workers. The sponsor-student relationship is the most obvious example of a model based on kinship roles; hence, we refer to scientists' "academic paternity" (in German doctoral sponsor is *Doktorvater*, or doctor-father). Some scientists construct elaborate kinship systems, tracing their academic lineage to famous scientific progenitors. Students from different cohorts who worked with the same sponsor may play mutually supportive roles, such as those of older and younger sib-

lings or cousins (while students in the same cohort may exhibit “sibling rivalry”). These links provide neophyte scientists with a ready-made set of extrainstitutional colleagues in their specialty and can be beneficial—as long as they do not prevent the development of normal collegial relationships. And that, of course, is the crux of the issue: Do sex roles borrowed from nonscientific arenas replace or distort the normal collegueship that is necessary for conducting scientific work? And, if they do, to what effect?

Consider the father-daughter model, in which a senior male scientist relates to a junior female paternalistically, and she responds with a combination of deference, docility, and affection. Scientists may use this model whenever an age or status difference exists, but it is most apparent in the relationships between women and their sponsors. In their broad outline, the relationships between sponsors and their protégés probably do not differ substantially by sex; both women and men profit from sponsorship in several ways. Sponsors both literally and figuratively place their students in the scientific stratification system and can intervene to provide a variety of professional opportunities. Active sponsorship also may provide young scientists with a ready-made academic extended family. These secondary connections with their sponsors’ other students may be especially valuable to female scientists, who might otherwise encounter obstacles to forming collegial ties, and probably represent a large proportion of women’s extrainstitutional linkages. Of course, some scientists of both sexes may dislike the paternalistic aspects of sponsorship (and women probably get larger doses of paternalism), but its advantages (Reskin, 1977b) usually outweigh any psychological costs.

The similarity in broad outline does not suggest an absence of sex differences in sponsorship. Just as father-son and father-daughter relationships differ in character, so too will relationships modeled after them. A male scientist and his sponsor both recognize that the former usually can expect eventually to be the latter’s equal—even if the student must rebel to achieve that status. This usually is not true of the relationships between most female students and their male sponsors. A long-term father-daughter association is perfectly consistent with the lower status that comes with being female, and women can get trapped in long-term pater-

nalistic relationships that discourage their scientific growth and independence. Despite the disadvantages that paternalistic sponsorship might hold, though, female scientists probably suffer more from the absence of sponsorship (Bayer, 1970, p. 13; Epstein, 1970b; American Association for the Advancement of Science, 1977) than from sex differentiation in its form.

Relationships based on the mother-son model are rare because few women scientists have access to graduate students (Bayer, 1973, p. 24), and fewer still are high enough in status to attract young male scientists. Nobel Laureate Rosalyn Yalow is among the few women scientists who successfully attracted younger male colleagues (whom she calls her "professional children"; Stone, 1978). Superficially, this model is benign (particularly for Nobel-caliber scientists), but Kanter (1977, p. 982) has noted some adverse effects on women's task performance. Female sales trainees in a predominantly male organization whose relations with male trainees were of this type were rewarded for maternal solicitude rather than independent action; as "accepting, good mothers" they could not criticize others; and they were defined as emotional rather than task specialists. If female scientists who relate to men via this role have higher status than their male role partners and are of demonstrated professional competence, these difficulties should be negligible. But if they are junior in status and are assigned to or assume the role on the basis of their age or personality, the problems that Kanter enumerates could occur.

Models Based on Marital Roles. As the most common adult sex roles, marital roles provide an obvious model for female-male relationships in other settings (Strodtbeck and Mann, 1956). Yalow's long-standing collaboration with the late Solomon Berson exemplified this pattern; although they were not married to each other, she and Berson have been likened to "an old married couple" (Stone, 1978). This model is most apt for researchers of approximately equal scientific status who are friends as well as co-workers. Thus far, because of the differences in the scientific statuses of female and male scientists, relatively few women have been professionally eligible for such a relationship. As the larger recent cohorts of female science students grow older, however, the number of such relationships may increase. Although some scien-

tists may shun a “professional marriage” in the belief that such an intimate working relationship may compromise them, its occurrence is favored by the presence of actual, and occasionally highly visible, husband-wife teams in the scientific community (three of the five female Nobel Laureates were joint recipients with their husbands). Most of the following discussion of the “professional marriage” model applies equally to scientists who are actually married to each other; I refer to the latter as “legal” spouses to distinguish them from “professional spouses,” whose relationship is only modeled after that of spouses.

Unlike parent-progeny and sibling roles, which are not gender-specific, marital roles prescribe a sexual division of labor and rewards. Hence, as models for relationships between female and male scientists, they may lead to task differentiation in which authority and creativity are reserved for the professional husband. No systematic evidence is available on sexual division of labor in relationships between professional spouses, but several of the female doctoral chemists whom I studied (Reskin, 1973), who were actually married to other chemists, worked as research associates in their husbands’ laboratories. Furthermore, for professional couples who also are legal spouses, any division of labor may well be compounded by differences in rank resulting from institutional antinepotism regulations.

Role differentiation may extend to areas other than scientific work. Yalow, for example, commented that Berson would have been very upset if she did not defer to him and recounted an occasion when they visited a scientist at his home and Berson advised her to “stay with the wives” (Stone, 1978, p. 98). What is important here is that he would ask and that, even today, the prevailing wisdom among women in male-dominated fields is to talk with co-workers’ wives at social gatherings, even though they may miss some professionally valuable conversations with fellow researchers.

Task differentiation on the basis of sex and marital roles will lead directly to differentiation in the allocation of credit for joint work.³ Because the professional husband is assumed to play the dominant role, most of the credit accrues to him. For example, although Nobel Laureate Maria Goeppert Mayer and her husband,

Joseph Mayer, jointly authored the influential *Statistical Mechanics*, many people assumed that it was largely his work (Dash, 1973, pp. 33–34). Among sociologists, men were disproportionately senior authors on papers authored with women, and the sex difference was more pronounced when the coauthor was their legal wife (Wilkie and Allen, 1975, p. 22). Of course, these differences may result from the men's (husbands') higher rank (itself a reflection of sex stratification⁴), but they also are consistent with sex differentiation in the distribution of recognition.

Introducing the roles of spouses into the research setting gives rise to yet another nonscientific model for relationships between male and female co-workers. The interaction between one member of a professional couple and her or his co-workers may center around the scientific interests of the former's spouse rather than her or his own work. For professional wives, who are more likely than professional husbands to experience this, the focus of the interaction with co-workers would be their husbands' work. Even if professional husbands occasionally have the same experience (usually in interactions with their professional wives' female colleagues), it will not hamper their collegial integration, because these interactions would constitute a small portion of their total professional contacts.⁵ Also, because the role of husband does not define men in the way that women are defined by being wives, occasional responses to husbands directed at their status as husbands do not preclude their simultaneously being regarded as colleagues.

The tendency for co-workers to interact with women who are either professionally or legally married to other scientists on the basis of their status as wives may lead such women to be superficially better integrated than unmarried women, but probably will not improve their chances of developing collegial relations based on their own professional interests or their status as scientists in their own right.

Regardless of their husbands' occupations, married female scientists apparently interact more frequently with male co-workers than do single women (Kaufman, 1978, pp. 16–17), perhaps because some men are more comfortable relating to married women (Bernard, 1964, pp. 198–199). For example, a senior male

academic commented on his close professional relationship with a younger woman in his department (Kaufman, 1978, p. 16): "We have worked on several projects together. I find her thorough, hardworking, and very insightful . . . I get along with her husband well too, and that makes for a good relationship as couples. That's helpful, too."

Possibly the best situation for a female scientist is marriage to a professional in another discipline. Her marital status would facilitate her social and professional integration, and the disciplinary difference would reduce the chance of her husband's receiving credit for her research contributions.

Models Based on Traditional Romantic Relationships. Social relationships characterized by some degree of sexual attraction provide additional models for cross-gender relationships between co-workers. These models, ranging from mild flirtation to love affairs, are derived from traditional patterns of romantic relationships; therefore, the behaviors involved are familiar to most adult women and men.

Flirtation is the most common of these alternatives, and for those who have trouble taking a female scientist seriously, it may make initial interaction easier. In keeping with general social patterns, women may flirt in exchange for the collegial attention they do not otherwise receive, or men may be more attentive to female co-workers who engage in some mild flirtation. At the other extreme are actual sexual relationships. Although gossip about real or suspected sexual liaisons is seldom absent when men and women work together, mentioning such relationships publicly, much less studying them systematically, has been taboo until so recently that evidence of their frequency does not yet exist. (For a theoretical discussion of this topic, see Bradford, Sargent, and Sprague, 1975.) Women scientists have only just begun to acknowledge these relationships and their structural implications—probably a first step to systematic study.

The use of traditional romantic relationships as models for cross-gender relations between co-workers adversely affects women scientists without providing any compensating advantages. For example, playing the role of girlfriend or lover usually does not improve social integration even superficially. In addition, some sci-

entists may find the roles of rational scientist and sexually attractive woman incompatible (Bernard, 1964, p. 198). Thus, women who are sexually attractive may have trouble shifting into the role of scientist, and some scientists may be unwilling to grant it to women who relate to male co-workers in terms of a traditional romantic model, even if the men themselves initiated that pattern of interaction.

Apart from the other disadvantages of these models, the sexual double standard holds the female partner responsible for romantic or sexual involvements with male co-workers. Although women customarily retain the right to accept or decline romantic overtures, male scientists probably have considerably more leeway in choosing which models to use and can avoid traditional romantic models altogether. Women, especially younger ones, may have more difficulty avoiding these models, and those who successfully eliminate all sexual or romantic aspects from their interaction do so at a price. They may try to play the mother or daughter roles discussed above (and accept the associated costs), or they may try to interact in a wholly professional way, but many of their associates will interpret this behavior as cold and unfeminine. Rosalind Franklin's appearance apparently evokes such a reaction from James Watson (Sayre, 1975, p. 21). Kanter presents a general discussion of the fate of the "iron maiden" in male-dominated organizations.

Because romantic roles depart from strict professional roles, even the suspicion of a sexual alliance casts doubt on whether women's professional achievements result from their own merit rather than from sexual favoritism (Rossi, 1973, p. 513). Consequently, even women who would never dream of sexual involvement with a co-worker monitor their behavior to avoid the appearance of trading on their sexual attractiveness. For example, to stop gossip by co-workers, a female informant had to quit traveling with her male assistants on research trips.

Some scientists believe that sexual attractiveness gives women a professional advantage and understandably resent the possibility that a co-worker will be rewarded on grounds other than performance, especially when the channels for those rewards are not available to them. However, the advantages that a few women

might obtain are small and fleeting; and, in the long run those advantages will prevent their being treated as serious colleagues or given full credit for their achievements. True collegial relations between persons who are unequal on some nonscientific status dimension can occur only if the discrepant status dimension is ignored altogether rather than emphasized through status-differentiated behavior.

Models Based on Quasi-Scientific Roles. Two additional models for relationships between female and male scientists—the scientist-technician and the scientist-apprentice models—can be considered together. In neither case are the roles gender specific, but women in research settings disproportionately hold the lower-status positions of technician or student. Because these roles are well known to all scientists and readily available as models for relationships between female and male researchers, they may be more acceptable than models based on sex roles. However, both models put women into subordinate roles incommensurate with their professional qualifications. (Professional women in other occupations also are vulnerable to role demotion, and some respond by remaining ignorant of the skills associated with the traditional female roles to which they might be demoted by co-workers; one example is the female manager who assiduously avoids learning to type.) No systematic evidence exists for the frequency of such models among female and male researchers with equivalent credentials, but anecdotal evidence documents their occurrence. For example, Nobel Prize winner James Watson (1968, pp. 20–21) apparently believed that Rosalind Franklin was co-Laureate Maurice Wilkins' assistant, although this was not the case and neither Wilkins nor the director of the Kings College Laboratory ever represented her as such (Sayre, 1975, p. 20). At least Watson was in good company, for Linus Pauling made the same mistake (Hubbard, 1976, p. 234). Sayre's (1975) account of Franklin's role in the discovery of the structure of DNA suggests that this misidentification helped Watson to justify using Franklin's crystallographic data without her knowledge or permission.

The features of female scientists' careers encourage the substitution of technician or student roles for that of colleague. First, women are disproportionately inbred at their Ph.D. departments

(Reskin, 1973), where they previously related to faculty in the professor-student model. Second, women doctorates in science who work at universities are disproportionately employed as research associates. Given the ambiguity of the research associate role, an incumbent—even one who holds a Ph. D. degree—is particularly likely to be defined as a technician rather than as a full professional. The roles of both student and technician are characterized by lower status and by a technical division of labor that allocates scientific creativity and decision making to scientists and laboratory work to those assigned the role of technician or student. Credit, presumably, is distributed accordingly. Both roles also foster female incumbents' structural dependence on higher-status males for direction and approval, thereby discouraging in the women the independence and creativity that are necessary for professional advancement.

Sex Differentiation and Scientific Processes

Space does not permit a full analysis of the implications of sex differentiation for women's participation in the social organization of science. Therefore, I have selected as illustrations two processes that are particularly crucial to the research role: informal communication and patterns of collaboration.

Informal Communication. The communication system in science includes all institutional arrangements and customs that affect the transmission of scientific messages among scientists (Menzel, 1962, p. 419). Upon discovering the importance to researchers of apparently accidental information, Menzel concluded that informal information networks actually constitute a highly organized communication system. Similarly, in a 1959 study Glass and Norwood (cited in Kjerulff and Blood, 1973, p. 625) found that casual conversation is the most important source of research information. Scientists use informal communication to supplement their knowledge of findings and to obtain technical information and procedural details unavailable in the literature. Colleagues provide information about unpublished work or about published work that a co-worker may not have read. Informal discussions also can reveal common interests, provide scientists with collegial reactions,

and may even lead to collaboration. They may also alert scientists to the risk of being anticipated and can help to ensure the priority of unpublished work.

A variety of professional (as distinguished from scientific) information also is transmitted informally: sources of materials, data, or services; tips about funding opportunities, potential employees, or job prospects; and news of the current research of other scientists. Although this information is sometimes available through formal channels, acquiring it can occupy most of a scientist's time. Moreover, some professional gossip may never become available outside informal networks. Missing out on stories about a colleague's squash game or prodigious drinking capacity will not hamper a scientist's performance, but careers can suffer if a scientist does not hear gossip about who is helpful and who is not to be trusted, or who is an operator and who is a "son of a bitch." Hagstrom (1965, p. 115) quotes an informant who was warned, on his first job, against working with an older man who exploited collaborators. Information about other scientists' moves, collaborations, or memberships on review panels also is useful.

Information flows into a local system through its members' contacts with outsiders and is transmitted within it during daily shoptalk in the labs and corridors and at lunch, in the locker room, over drinks, or during the evening while their families socialize. Scientists not present for such conversations miss out on their content. Conversations held away from the work setting obviously exclude persons not involved in such activities.

The exclusion of women from conventional collegial relationships, the substitution of gender-based role relations for collegial roles, and other forms of sex-role differentiation all reduce women's access to informal communication. Sex-role differentiation will reduce the probability that male scientists will define a woman as an interested and trustworthy recipient for various types of information (Hughes, 1945, p. 356) and even the likelihood that she will be present when information is being exchanged. Scientists may not remember to tell women about funding opportunities or special conferences if the women's roles with male co-workers are modeled after that of the student or technician. If it occurs to them

at all, they probably assume that women with close ties to a particular man will receive the information from him. Even if that assumption is sound, forgetfulness and delays in the transfer of information may put the woman at a disadvantage. The ultimate result is that any marginality in women's positions in informal communication systems is self-perpetuating: women receive less information and hence lack information to exchange, and so on. Also, women's assignment to or acceptance of social-emotional roles such as the mother role could lead them to specialize in personal rather than professional information, thereby reinforcing their image as oriented toward people rather than the profession.

Extraintitutional communication flows between former students and teachers, former co-workers, and scientists who became acquainted at professional meetings, during laboratory visits, or through correspondence. However, because female faculty teach fewer graduate students (Patterson, 1971; Bayer, 1973, p. 24), they have fewer former students to transmit information. Because their ties with their academic sponsors are weaker (Roe, 1966; Bayer, 1970, p. 13), they are less likely to be in touch with them. If their relationships with former co-workers were not of the traditional collegial variety, they probably did not persist when one party moved. When it comes to contacts outside their institution (Bayer, 1970, p. 18), female scientists are especially disadvantaged. Female bioscientists received fewer invitations to participate in events away from their labs (Bernard, 1964; Bayer, 1973, p. 28) and received fewer preprints from scientists at other institutions (Bernard, 1964).

When women do interact with noninstitutional colleagues at meetings, their interactions may follow traditional sex roles (Mintz, 1967) and thus have little professional value. Female bioscientists found discussions at meetings a less important source of scientific information than did males, and women zoologists were substantially less likely to report that their most productive conversations were with persons whose work was just becoming known (Bernard, 1964). Because scientists find informal conversations more useful than the formal presentations (Kaplan and Storer, 1968, p. 115), women's restricted access to such conversations could explain these differences. Male scientists who are unwilling to extend colleague-

ship to a woman also will be constrained in dealing with female scientists whom they do not know and may refrain from approaching a woman after she has given a formal presentation. If male scientists typically hesitate about such contacts, then women will lack feedback on presentations that other scientists find extremely valuable (Lin, Garvey, and Nelson, 1970, pp. 30–31).

Scientists who are excluded from informal communication networks depend on less efficient and more time-consuming formal channels of information. For example, young people—presumably not yet well integrated in informal communication networks—were overrepresented among users of an innovative system that distributed unpublished material (Griffith and Miller, 1970, p. 135). Women bioscientists, predictably, depended more than men on published literature (Kaplan and Storer, 1968, p. 115).

Unlike formal communication, informal communication is a two-way process that helps researchers establish the relevance of others' results for their own work, direct the communication process to their particular needs, speculate about findings, and obtain immediate feedback (Garvey and Griffith, 1967, p. 1013). But two-way communication is less likely to occur between female and male co-workers whose relationship is modeled after roles that assign initiative to the male and define the female as the receiver of messages (information, directions, and so forth; see Mintz's 1967 account of the typical communication between female and male sociologists at their annual professional meetings). For women, then, sex differentiation reduces the value of informal communication by reducing the likelihood of two-way communication.

Collaboration. Patterns of collaboration among scientists reflect work organization, scientific stratification, specialization, the cognitive structure of specialties, and informal collegial relations as well as the preference of individual scientists for joint or solo work. Because collaboration patterns are influenced by both formal and informal aspects of the social organization of science, they too will reflect sex differentiation in the scientific community. Hagstrom (1965) provides several insights that suggest hypotheses about sex differentiation in collaboration. Because collaboration usually begins informally, anything that affects the frequency of informal contacts between colleagues will affect the likelihood of collabora-

tion (Hagstrom, 1965, p. 114 and p. 122, table 12). Scientists broach the idea of collaboration cautiously, particularly if the suggestion might imply their inability to solve a problem independently. They can reduce the risk of being turned down by tentatively sounding out a potential collaborator in advance, but the possibility of rejection probably deters individuals who are particularly vulnerable. For this reason and others, collaboration tends to occur among equals (Hagstrom, 1965).

Sex-role differentiation in science may affect female scientists' opportunity to collaborate. If it restricts their participation in informal communications systems, it will reduce their access to potential collaborators. Certain borrowed models of relationships will hinder collaboration. Women in explicitly lower-status positions than those of their male role partners (the apprentice, technician, or daughter roles) are more vulnerable to rejection and should be more reluctant to propose collaboration. Although male scientists are not constrained from initiating collaboration, the tendency for collaborators to be peers should lead joint efforts among unequals to exhibit a substantial division of labor by gender, assigning decision making and creative tasks to the male. Moreover, if the women's lower status in such relationships is generally known, other co-workers might refrain from proposing collaborative work. To propose collaboration might imply their own professional inferiority to the women (and, conceivably, by transitivity, to her male role partners as well).

Roles that link female scientists with particular male colleagues also might discourage other scientists from viewing them as potential collaborators, especially if exclusive access is implied in the borrowed roles as defined in the larger society.⁶ If this reasoning is correct, proportionately more women than men should collaborate primarily with a single colleague rather than with several different collaborators. Female researchers who are legally or professionally married to other scientists should be freer than other women to initiate collaboration, both with their professional spouses and with other male scientists.

Hagstrom (1965, p. 118) suggests that scientists are less reticent about asking someone with obvious expertise to collaborate, because the request does not imply their own deficiency. So women

who are clearly experts in their specialties may have more opportunities to collaborate (Hagstrom, 1965, p. 124; Epstein, 1970b, p. 979). However, since these collaborative relationships would not be between equals, they probably would be limited to the duration of the project, and the interaction they involved might be relatively formal. For these reasons they would not necessarily improve the female researcher's integration into the collegial system or even increase her future chances to collaborate. In fact, they may help to define her as a technical expert rather than a full colleague.

Other variables also may reduce collaboration between the sexes. Scientists who believe that male-female working relationships have a sexual component would avoid working with women lest others misinterpret the association. Although in jest, Yalow's dissertation director explained that he accepted her as a graduate student because he was so old that his motives would not be suspect (Stone, 1978, p. 30), thus illustrating how near the surface such concerns often are. Some scientists may simply feel uncomfortable working with members of the opposite sex. Male researchers might have such feelings for a whole host of reasons related to women's discrepant gender status. Women may not pursue or may even avoid opportunities to collaborate with men because they fear—with some reason—that they will not receive full credit for their contributions. Women who work with their legal husbands are particularly vulnerable in this regard; even first authorship for them may be discounted as an expression of conjugal affection (Dash, 1973, p. 289). The evidence on sex differences in frequency of collaboration is largely anecdotal, though recent data show that female sociologists are less likely than males to publish collaborative papers (Chubin, 1974, p. 85; Mackie, 1976, p. 285). The magnitude of the differences should vary with the proportion of workers in a field who are female, but data from other disciplines are not yet available.

Indirect evidence of sex differentiation in science should exist in patterns of name order on coauthored papers. Conventionally, first authorship denotes greater contribution to the research and customarily goes to the scientist who initiated the collaboration (Hagstrom, 1965, p. 117), although it also may mirror differences in the academic statuses of the authors. The order in

which scientists' names appear on multiauthored papers is of consequence because recognition and other rewards often depend on the relative contribution that name order denotes (Zuckerman, 1968). If sex differentiation discourages women from proposing collaborations, it would indirectly reduce their likelihood of being senior author on joint papers (Wilkie and Allen, 1975, p. 22). The specific roles adapted for female-male relationships between collaborators also may affect the division of responsibility and, thus, credit for joint work. Both the scientist-technician and the professor-student relationships, which the sexes may use as models, routinely assign first authorship to the higher-status party—invariably the male. Men in these positions have the prerogative of dictating a different name order, but they also have the prerogative of omitting the names of junior collaborators. Mackie (1976, p. 289) speculates that female collaborators, like graduate students or technicians, may sometimes receive acknowledgments rather than coauthorship in return for their contributions. In addition, the division of labor congruent with borrowed roles may discourage women from contributing to the more creative aspects of the joint ventures, in which case they would not merit first authorship.

Sex Differentiation and Women's Scientific Performance

Sex-based functional differentiation in science has been well documented. Female and male scientists tend to work in different kinds of organizations and, within those organizations, typically fill different positions and perform different functions (Vetter and Babco, 1975; Gilford and Snyder, 1977). Some observers have suggested that such gender-based functional differentiation accounts in part for women's lower productivity. (Solmon, 1976, p. 24; Astin, 1977, p. 5). Sex differences are reduced though not eliminated by controlling for employment sector (Centra, 1974, pp. 68–72). More detailed examination of the actual roles female and male scientists occupy should further reduce these differentials. Consider the results of a survey of college and university faculty by the American Council on Education (Bayer, 1970, 1973). Female university faculty members spent more time than males teaching and advising students and correspondingly less time in administration and research. They taught more classes (including more intro-

ductory courses), were in class more hours per week, and not surprisingly were less satisfied with their teaching load. Although they taught more undergraduates, they were less likely to have teaching assistants. They also taught fewer graduate students and had fewer research assistants than did male faculty. That substantially more men than women held Ph.D. degrees accounts for some of these differences, and comparisons within fields should reduce the differences still further. Unfortunately, available information on the principal activities of university scientists is not presented separately for both field and sex (Vetter, 1977, pp. 24–25). But the differences reviewed here probably do not stem entirely from disciplinary differences. More likely they reflect real sex differences in role obligations within the sciences.

It is easy to see how these differences could affect performance of the research role. Undergraduate teaching takes time away from research without providing an opportunity to recruit students or get feedback on one's work. Bayer (1970) reports that university faculty women are more likely than men to agree that they hardly ever have time to give a piece of work the attention it deserves.

Teaching graduate students in one's area facilitates research. Seminars provide an opportunity for researchers to discuss their ongoing work with an intelligent captive audience and to attract students to a research project. If women teach fewer graduate students, they will find it more difficult to recruit them. In the experimental sciences, where research is a team effort, having fewer students presumably reduces women's research output. Hagstrom (1965, p. 120) found a negative correlation between the number of collaborators that scientists had and their number of graduate students (and both are correlates of productivity). Scientists with weaker collegial ties can compensate by working with a large group of students. Ironically, women apparently do not have this option; both their sex and their isolation from other members of the department probably deter the better students from working with them.

As mentioned, collegial ties and informal communication also are essential for scientific research (Crane, 1969; Mullins, 1973; Hargens, Mullins, and Hecht, 1977). Hagstrom (1965, p. 50)

reports that the number of papers his informants published in the previous three years was moderately related to their communications with departmental colleagues and strongly related to communications with disciplinary colleagues at other departments. Pelz (1956) found that the most productive medical researchers working in government typically had one close colleague with similar values and had frequent contacts with several additional colleagues. If collegueship is causally prior, then these associations point to obstacles to women's scientific performance. Lack of communication with colleagues limits access to necessary information, and women are more poorly integrated into informal communication systems. Thus, the "lucky accidents" that male biochemists experienced in their research were more likely to result from informal communication than those the women experienced. (Bernard, 1964, p. 273). Restricted communication also reduces scientists' motivation to do research (Hagstrom, 1965, p. 48–49), partly because poorly integrated scientists receive less encouragement and fewer informal rewards for their achievements.⁷ The stronger impact of formal citations on the productivity of women chemists (Reskin, 1978) is consistent with their receiving less informal recognition. Colleagues' expectations should be especially important to researchers who face conflicting demands from other institutional roles and rewards for nonresearch performance, and gender-based functional differentiation in the scientific community concentrates women in just such situations. Unfortunately, co-workers who do not expect women to be productive are unlikely to provide encouragement and informal rewards, thereby contributing to the fulfillment of their expectations.

Zuckerman and Merton (1972, p. 323) suggest that retention of the research role by older scientists depends partly on their having active researchers as their reference group. This applies equally to women, whose isolation may lead them to form reference groups among people outside their discipline or among nonscientists (teachers, administrators, or other professional women). Kaufman (1978, p. 17) found that female faculty in human ecology were more likely than male co-workers to name as members of their collegial network persons whose research interests differed widely from their own. One is hardly surprised, then, to learn that

these women were less likely than the men to find their colleague-friends professionally valuable.

Sex differentiation in collegial networks will increase the likelihood that women will be anticipated in publishing their research. Scientists are not anticipated by institutional co-workers, who know each other's work, but by people at different institutions. Researchers who learn through the grapevine that they are working on the same problem can forestall anticipation by agreeing to collaborate rather than compete (Hagstrom, 1965, p. 115), but collaboration is less available to female scientists. More poorly integrated into informal communication systems, they are less likely to learn of others who are working on the same problems. If they do learn of potential competitors, they are less likely to know them than male researchers would be, because they are more isolated from noninstitutional colleagues in their specialty. Finally, women probably would be more hesitant than men about proposing collaboration with scientists whom they do not know, because they are unsure of their reception (Epstein, 1970b, p. 976). As a result, women who are active researchers probably face a greater risk of being anticipated than similarly situated men do.

Summary and Conclusions

The collegial role prescribes a set of reciprocal obligations between researchers of approximately equal status, involving the exchange of information, encouragement, evaluation, and informal recognition. Without these collegial supports, performance of the scientific role is almost impossible. Because pronounced discrepancies in the scientific or social statuses of co-workers are inconsistent with collegueship, professional and technical workers of unequal scientific status relate in terms of other institutionalized roles. If male and female scientific co-workers do not both choose to ignore the status discrepancy introduced by gender, they cannot interact as full colleagues. Alternatively, they may minimize their interaction, thereby isolating female scientists, or they may draw on nonscientific role relationships that are consistent with their gender statuses but inconsistent, to varying degrees, with the scientific role. Either result reduces the likelihood that women will be fully integrated into collegial relations. In short, obstacles to collegiality be-

tween the sexes help to introduce sex differentiation into the scientific community. Its presence, in turn, will be manifest in the positions that the sexes typically occupy and in differences in their scientific role performance.

The evidence I have offered for this thesis is primarily anecdotal. Indeed, the argument itself is largely conjectural and should profit from the insights of other observers of the scientific community. Ultimately, a variety of factors that I could not consider here—factors such as the proportion of women in a discipline, its size and age, and its level of consensus or codification (Zuckerman and Merton, 1972)—may specify the processes of sex differentiation. But, regardless of whether each claim stands or falls, the foregoing should demonstrate the value and indeed the urgent need to go beyond simple questions about the extent of sex differences in science to examine how the social organization of scientific work and the research role foster sex differentiation in science.

Notes

1. Despite the importance of collegiality, analyses of the role of scientific researcher usually have ignored collegial obligations, probably because most scientists learn the collegial role without formal instruction and perform it automatically. Individually, we probably attribute poor collegueship to idiosyncratic factors such as “personality,” especially if the research role is performed adequately in other respects.

2. Were collegial roles to evolve in some new, sexually integrated profession, we could determine the importance of tradition relative to barriers resulting from the nature of collegueship (although in sexually integrated professions gender also should be less salient; see Kanter, 1977). But an increased proportion of women in a profession should also lead to various mechanisms that would mitigate the strain engendered by mutually exclusive gender and collegial roles. Segregation (Hughes, 1945, p. 358; Epstein, 1970a, p. 187) and functional differentiation are two possibilities. A third—substantially improved collegial integration of female scientists—seems less likely, given the apparent incompatibility between collegial equality and gender stratification.

3. This situation holds for all borrowed roles that assign women an inferior status; indeed, it is consistent with sex-role differentiation in general.

4. Wives often hold lower professional status than their husbands for two reasons. First, in keeping with social norms that husbands should hold higher status than their wives (an arrangement that reinforces men's gender-based authority and affords their spouses' apparent upward mobility), scientists may seek partners who will conform to the expectation. (Marriages between female graduate students and male faculty, which are favored by both proximity and the social norm, ensure the husbands' professional superiority.) Second, among married couples who initially held equal professional status, the husbands may advance more rapidly for a variety of reasons, some of which are discussed in this chapter.

5. An exception would be a man married to a female scientist of such high status that most of his colleagues' interaction with him would focus primarily on her work rather than his own. Husbands of very successful women in professions such as entertainment, sports, or politics also experience this and may assume some semiprofessional role directing their wives' careers. Such a role both legitimizes their interacting with others in terms of their wives' status and permits the appearance of dominance. This option is not available to men trained in research, and the relationships of scientific couples in this situation may be strained.

6. Scientists' wariness about appropriate behavior toward their colleagues' spouses when the latter are also scientists is reflected in their tendency to accompany professional contacts with jokes about jealous spouses.

7. Epstein (1973, p. 68) has suggested that women may be rewarded for merely adequate performance, because some men find *any* performance by a woman commendable, or that they may even be rewarded for irrelevant performance. Of course, rewards that are not contingent on performance do not encourage professional achievements.

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